

by typing the URL <http://127.0.0.1/moodle/login/index.php>.

Now login as the user *Shishya* to Moodle. Go to the course page of BBB101; you will see a link similar to the one shown in Figure 3. Click on this link and press the button 'Join session'; now both *Guru* and *Shishya* are part of the Web conference we had set up earlier. Figure 4 shows one of the authors (Deepu) logged in as both *Guru* and *Shishya*. Move your hand in front of the webcam and you will see one of the videos trailing behind the other, proving that this indeed is a Web conference.

Now, we can test some of the features of BigBlueButton.

BigBlueButton offers a lot of features to make online teaching effective, like note sharing, computer screen sharing, shared virtual whiteboards, breakout rooms, public chats, etc. Note sharing allows a teacher to share notes prepared to the students during the live Web conference. Computer screen sharing allows the teacher to show Web pages and other documents opened in his or her computer to the students, if required. Shared virtual whiteboards allow people to perform collaborative work by drawing and writing on a single white screen.

'Breakout rooms' is another important feature provided by the BigBlueButton. Assume you are a teacher in a class with 50 students. In a real classroom, often these students can be divided into say, 10 groups of 5 members each, so that they can be assigned with problems to work on. Breakout rooms allow us to simulate the same in virtual classrooms. BigBlueButton allows the creation of groups so that communication is possible only between the members of a group and not the entire class.

Public chat is also quite useful as students can pose questions during live tutorial sessions. Figure 5 shows a public chat between *Guru* and *Shishya*. Public chat is one of the few features of BigBlueButton available to a user

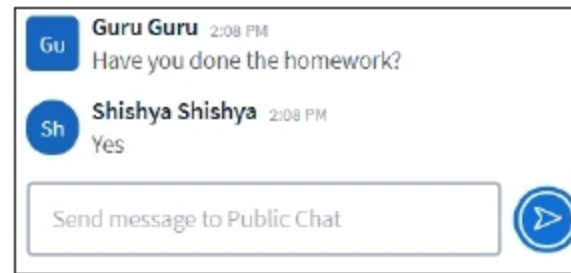


Figure 5: Public chat in BigBlueButton

assigned with the role of a viewer. Though the chat in Figure 5 looks simple, notice that you have to send messages from the two different logins to get this effect, not just from a single login.

BigBlueButton also allows you to conduct instant polls so that the presence and attentiveness of participants can be assessed. If a teacher conducts random polls during online sessions, then no student will have the courage to attend to matters of lesser importance -- say, play some online game. BigBlueButton also allows you to record the sessions so that participants can review them later. It allows you to share your presentations with the participants.

An essential part of online courses is the ability to play pre-recorded videos and audios. This is one feature where the use of BigBlueButton makes your day a lot easier. With BigBlueButton, it is very easy to play pre-recorded videos and audios in the middle of live online sessions. On the bottom left corner of the BigBlueButton window, you will see a plus symbol (+). Click on it and you will

get the following three options to choose from: start a poll, upload a presentation, and share an external video.

Click on the option 'share an external video'. You will see a window popping up with a text-box to enter the external video link. In this text-box, copy and paste the link to the video you want to show during the online session and click on the button 'Share a new video'. The participants will be able to see the video on the presentation area of BigBlueButton. Figure 6 shows the playback of a video about OSFY we found on YouTube.

Now that we tested our BigBlueButtonBN plugin in the local copy of Moodle, let us go to the next level. How can we use BigBlueButtonBN plugin in a real scenario? For this, we need to host a BigBlueButton server. One of the authors (Manoj) hosted a BigBlueButton server to run a real remote Web conference, not just in the two windows of a Web browser in our own systems. Figure 7 shows a real Web conference between *Guru* and *Shishya*, from Kottayam and Kannur in Kerala, approximately 350 kilometres apart.

Since our aim is to introduce the BigBlueButtonBN plugin and list out its features, we refrain from including the details for setting up the BigBlueButton server. An easy option is to use a shell script called *bbb-install.sh* to automate the installation

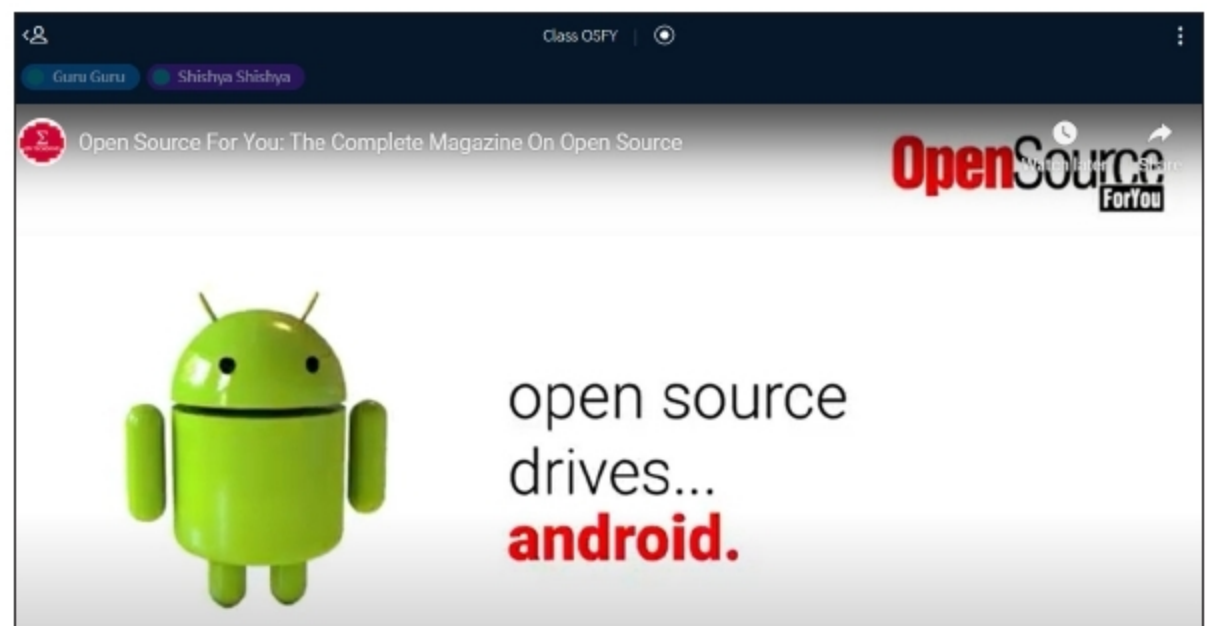


Figure 6: Video playback in BigBlueButton